

Passwords are a fundamental part of cyber security, used to control access to personal accounts, systems, and sensitive information. Despite their importance, they are often one of the weakest points in security due to poor practices such as reuse, simplicity, or improper storage. Understanding how passwords are used, how they are attacked, and how to manage them effectively is essential for both individuals and organisations.

At a basic level, a password is intended to verify identity. However, attackers use a variety of techniques to bypass or compromise passwords. These include guessing common passwords, using automated tools to try large numbers of combinations, or exploiting data breaches where passwords have already been exposed. In many cases, weak or reused passwords allow attackers to gain access with minimal effort.

There are several common types of password-related attacks:

- **Brute force attacks** – automated attempts to guess passwords through repeated combinations
- **Dictionary attacks** – using lists of common words and passwords
- **Credential stuffing** – using previously leaked usernames and passwords on multiple sites
- **Password spraying** – trying a small number of common passwords across many accounts
- **Phishing for credentials** – tricking users into revealing passwords through deceptive messages

These methods are effective because many users rely on predictable or reused passwords across different services.

For individuals, compromised passwords can lead to unauthorised access to email, banking, and social media accounts, often resulting in financial loss or identity theft. Because many services are linked through a single email account, gaining access to one password can allow attackers to reset and take control of others.

For businesses, password weaknesses can result in unauthorised access to internal systems, sensitive data breaches, and wider network compromise. A single compromised account—particularly one with elevated privileges—can have significant consequences for operations and security.

Recognising poor password practices is an important step in reducing risk. Common issues include:

- Using short or simple passwords that are easy to guess
- Reusing the same password across multiple accounts

- Sharing passwords between users
- Storing passwords in unsecured locations
- Using predictable patterns such as names or common words

These behaviours increase the likelihood that passwords can be guessed, stolen, or reused by attackers.

Improving password security involves adopting stronger practices and supporting technologies:

- Creating long, complex, and unique passwords for each account
- Using passphrases made up of multiple unrelated words
- Enabling multi-factor authentication (MFA) to provide an additional layer of security
- Using password managers to securely store and generate passwords
- Regularly updating passwords, particularly after a known breach
- Avoiding sharing credentials or storing them in plain text

For organisations, additional controls such as enforced password policies, account lockout mechanisms, and user access management can further strengthen security.

Passwords remain a critical component of access control, but they are most effective when combined with other security measures. As cyber threats continue to evolve, relying solely on passwords is no longer sufficient, making it important to adopt layered approaches to authentication.

This page provides an overview of password security, including common risks, methods of attack, and ways to improve protection. Developing strong password practices is a key step in reducing vulnerability and maintaining secure access to systems and information.